

DATA DETECTIVES



Key

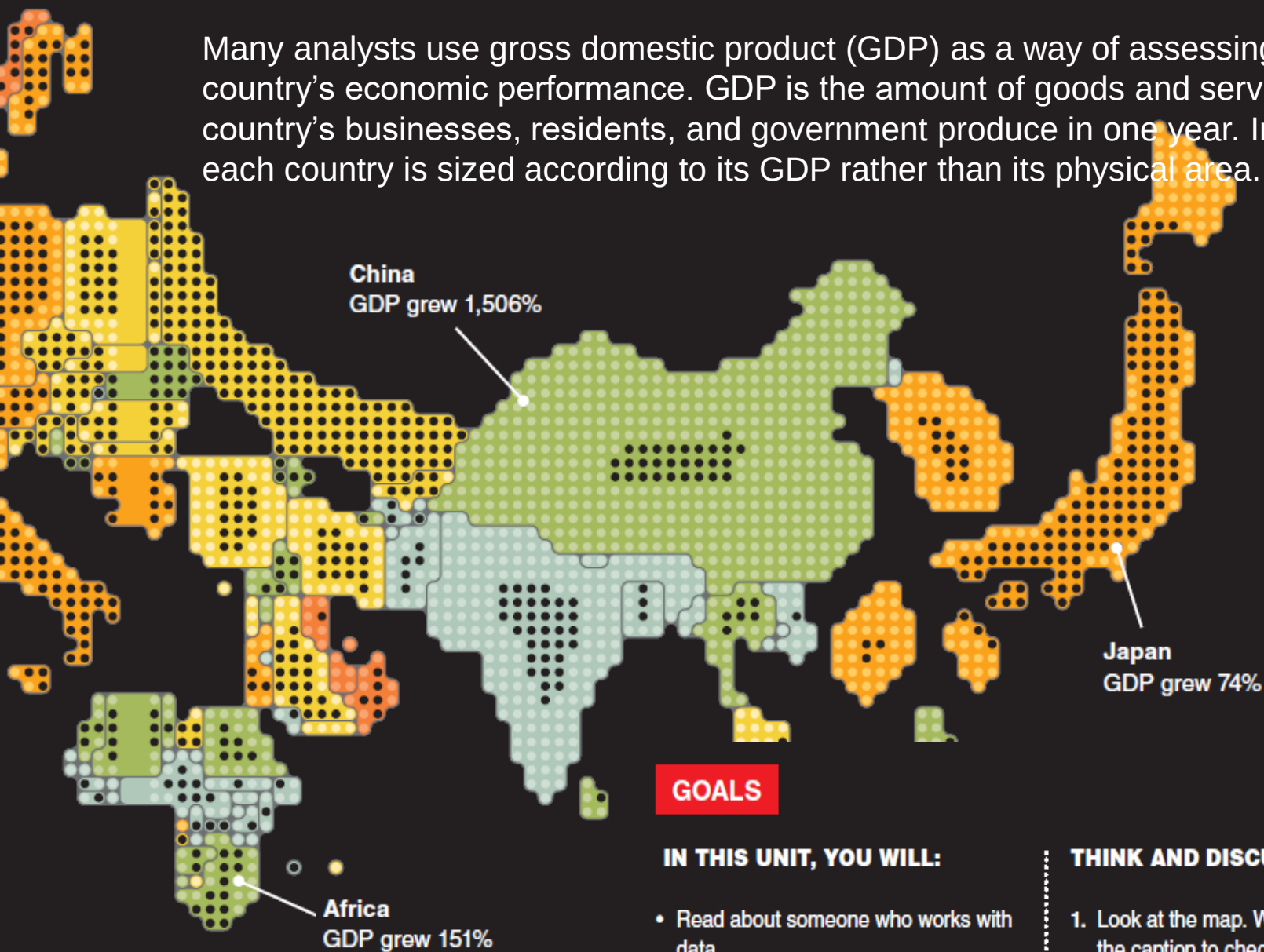
- 1980 GDP (black dots)
- 2009 GDP (all dots)

Color indicates gross domestic product per person (2009).

- More than \$40,000
- \$25,000 to \$40,000
- \$9,514 to \$24,999
- \$3,000 to \$9,513
- Less than \$3,000

World average:
\$9,514

Many analysts use gross domestic product (GDP) as a way of assessing a country's economic performance. GDP is the amount of goods and services that a country's businesses, residents, and government produce in one year. In this map, each country is sized according to its GDP rather than its physical area.



GOALS

IN THIS UNIT, YOU WILL:

- Read about someone who works with data.
- Learn about the benefits of visual data.
- Explore your own social data.

THINK AND DISCUSS

1. Look at the map. What does it show? Read the caption to check your answer.
2. What are the advantages of showing data in this way?

Lesson A

PRE-READING

A. What does the infographic on these pages show? Choose the best three answers.

- 1. Selected cities' heights above sea level
- 2. Predicted height of sea levels over time
- 3. When selected cities will be below sea level
- 4. Why ice sheets are getting smaller and sea levels are rising

B. What message, or story, does this infographic tell? Write your ideas and then discuss with a partner.

It shows how and when rising sea levels, due to the melting of ice sheets, will cover major world cities.

- C.** Look at the Venn diagram and caption on page 138. Do you think the “When Sea Levels Attack!” infographic is an example of effective information design? Why or why not? Discuss with a partner.

Answers may vary, but most will likely agree that the infographic is interesting and accurate while offering an attractive design that's easy to read.

- D.** Read the introductory paragraph on page 137. Why do you think David McCandless thinks information is beautiful? Discuss your ideas with a partner.

Answers will vary. McCandless explains that the hidden connections, patterns, and stories in information is what makes it beautiful. This is what he tries to show in his infographics.

years sea level



8000 80m

When Sea Levels Attack!

How long have we got?

1000 20m

8m

7m

400 6m

5m

300 4m

200 3m

2m

100 1m

Venice

Amsterdam

Hamburg

Saint Petersburg

Los Angeles (Seafront)

San Francisco

Lower Manhattan

New Orleans (lowest levee)

South London

Shanghai

Edinburgh

New York

London

Taipei

800 years



80 years



TOTAL CONTRIBUTIONS

Antarctic ice sheet (South Pole)
73m

Greenland ice sheet
6.5m

All other ice (Arctic included)
0.5m

by 2100, worst case scenario: 1m

Already happened since 1870
20cm

Note: Heights above sea level vary across cities. Lowest points used.

Source: IPCC, NASA, Realclimate. org, NewScientist.co, Potsdam Institute, Sea Level Explorer



INFORMATION IS BEAUTIFUL

With the vast amount of data available today, some of us may be suffering from data glut. Journalist David McCandless thinks that the solution lies in well-designed visuals that can help us make sense of a **complex** world.

“There’s something almost quite **magical** about visual information,” says David McCandless. He and his team use software to scrape—collect—data from lots of different sources online. They try to find hidden **connections**, patterns, and stories that connect the pieces of data. Then they **create** infographics that people might find interesting and useful. Also, as McCandless says, infographics can “just look cool!”



For example, a world map usually represents the size of countries by land area. But what does the world look like if we see nations according to their population size, or how much they **consume**? This kind of visual not only helps us understand the world better; it can also change our perspective about it. By connecting different data, we can get a more complete picture of the world.

According to McCandless, by **visualizing** information, we turn it into a landscape that we can explore with our eyes—“a sort of information map.” As McCandless explains, the human eye easily notices and **appreciates** visual patterns. McCandless calls patterns the “language of the eye.” On the other hand, words and numbers are the “language of the mind.” McCandless believes that if you combine patterns with words and numbers, you can speak two languages at the same time, “each **enhancing** the other.”



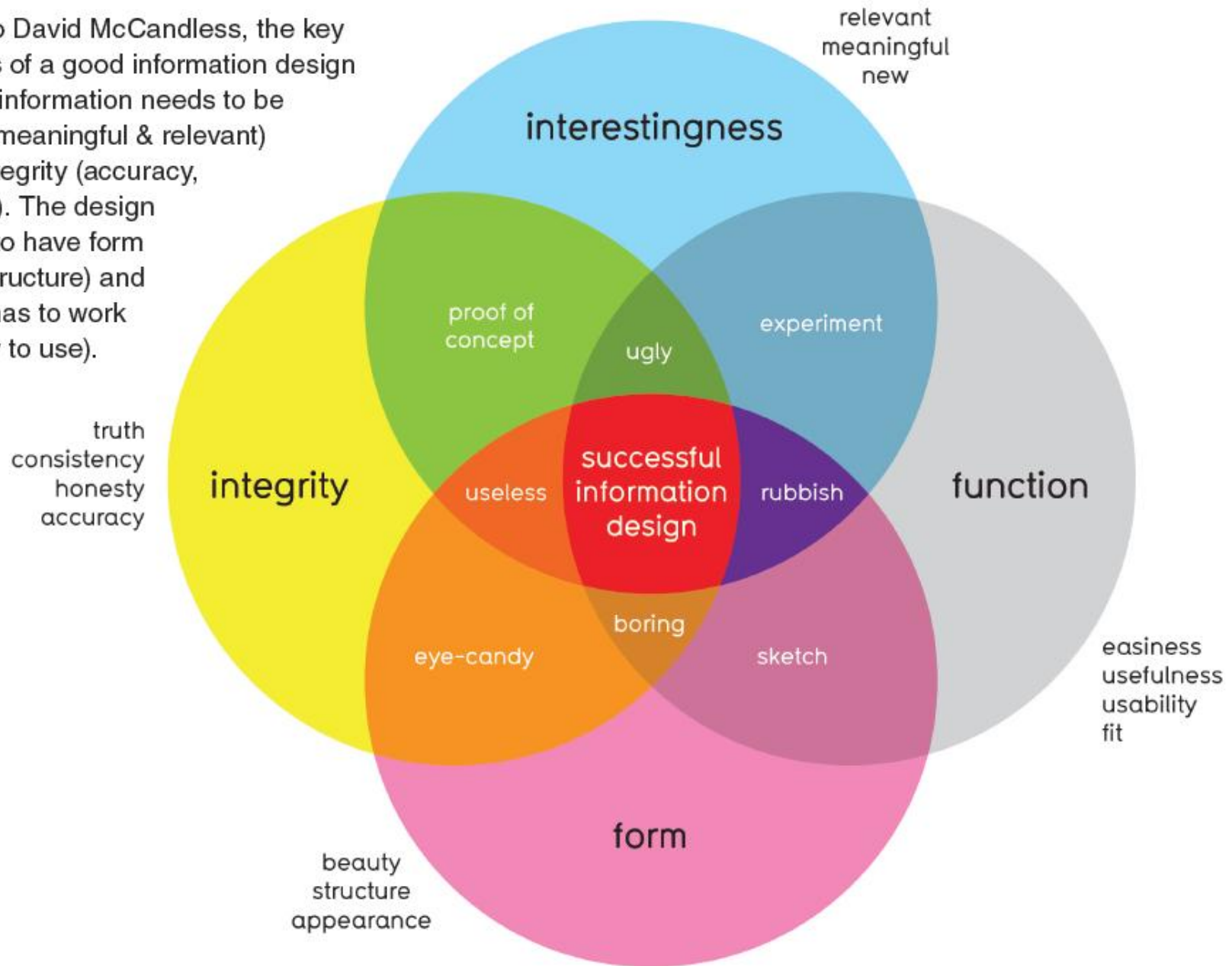
The solution to information overload, according to McCandless, is using our eyes more. By visualizing information, we can understand it better. McCandless compares the act of reading a large amount of data to the experience of finding your way through the wilderness. When you come across a beautiful graphic, he says, it's a relief. "It's like coming across a clearing in the jungle."

glut: *n.* too much of something that cannot all be used

wilderness: *n.* an area of natural land unused by people

What Makes Good Information Design?

According to David McCandless, the key components of a good information design are that the information needs to be interesting (meaningful & relevant) and have integrity (accuracy, consistency). The design also needs to have form (beauty & structure) and function (it has to work and be easy to use).



Developing Reading Skills

GETTING THE MAIN IDEAS

Use information from the passage on pages 136–138 to answer each question. Choose the best answer.

1. What does David McCandless do?
 - a. He studies visual data and explains it with words.
 - ☒ b. He collects data and presents it visually.
 - c. He reviews visual data for large organizations.
2. What problem does McCandless want to solve?
 - a. We don't get enough data every day.
 - ☒ b. We get too much data every day.
 - c. We ignore online data unless it is in an infographic.

UNDERSTANDING DETAILS

A. Why is presenting information visually useful? Choose the four ideas that are stated in the passage.

Infographics help us understand information better because they _____.

- 1. can be understood by people who speak different languages
- 2. tell a story
- 3. change our perspective
- 4. connect different sources of data together
- 5. let us focus only on the important information
- 6. are more entertaining to look at
- 7. give us a more complete picture of the world
- 8. take less time to read than words and numbers

- B.** Look at the choices in Exercise A that are not mentioned in the reading. Do you think that these are also examples of the benefits of visual data? Discuss with a partner.

Answers will vary.

- C.** What do McCandless and his team do when they look at different sources of data? Complete each sentence.

- a. They gather together *data from various sources*.
- b. They try to discover *connections, patterns, and stories within data*
- c. They produce *interesting and useful infographics*.

UNDERSTANDING INFOGRAPHICS

Many people use infographics to tell a story or explain complex information visually. Most infographics include a mixture of text, pictures, and statistics. Being able to read infographics is an important 21st-century skill.

Use information from the infographic on pages 134–135 to answer each question. Discuss your answers with a partner.

1. What do the size and color of each country represent?

The size represents the country's GDP relative to others, while the color shows the average GDP per person in that country.

2. What do the black dots and the light dots represent?

The black dots show the size of the GDP in 1980, while all the dots (light and dark) show the GDP in 2009.

3. Circle the region whose GDP has grown the most since 1980.
- a. United States
 - b. South America
 - c. China
4. McCandless calls patterns the “language of the eye” and words and numbers the “language of the mind.” How do you think this infographic uses both the language of the eye and the language of the mind? Discuss with a partner.

The map and the dots are the graphic representations of the information, while the key and the notes on the map help explain it.

GETTING MEANING FROM CONTEXT

- A.** The passage states that we are suffering from “data glut.” What does this phrase mean? Write your idea and discuss with a partner.

“Data glut” refers to the overwhelming amount of data accessible to us today

- B.** Find and circle the two-word expression in the passage that has the same meaning as data *glut*.

information overload (the first line of Paragraph 4)

BUILDING VOCABULARY

A. Choose the word or phrase that has the closest meaning to the bolded word.

b 1. **magical**

a. difficult

b. amazing

c. complete

c 2. **enhancing**

a. confusing the issue

b. changing the style

c. improving the quality

a 3. **consume**

a. use

b. like

c. want

a 4. **complex**

a. hard to understand

b. easy to understand

c. important to understand

B. Use the “Bandwidth of the Senses” infographic and the words below to complete the paragraph.

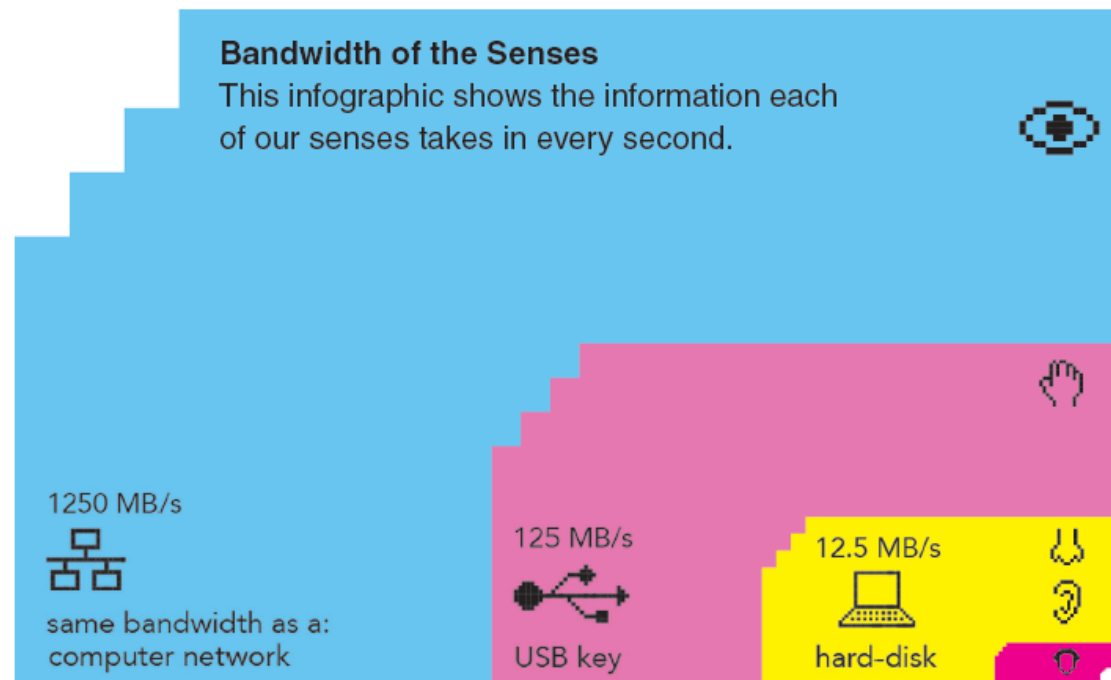
appreciate

connection

created

visualize

Danish physicist Tor Norretranders created this infographic to help us visualize the power of our senses: sight, touch, smell, hearing, and taste. Norretranders made a(n) connection between our senses and computing equipment to show how quickly we process information with each of our senses. Our sense of sight is the fastest. It's like a computer network. Taste is the slowest. It processes information as slowly as a pocket calculator. This infographic helps us better appreciate how powerful our sense of sight is in comparison with our other senses. Most importantly, however, the tiny white section in the bottom right corner shows how much information we are actually aware of: just 0.7 percent of what we see, smell, taste, touch, and hear.



CRITICAL THINKING

1. **Inferring.** McCandless says that when you come across a beautiful graphic, “it’s like coming across a clearing in the jungle.” What do you think he means by this?

Since we read so much of the information we take in, an infographic gives our eyes and brain a break, and a new way of looking at things.

2. **Personalizing.** Think about how you spend a typical day. What do you do? How long do you do each thing? What kind of infographic do you think might best show this data? Make an infographic about your day and share it with a partner.

Answers will vary.

EXPLORE MORE

Go to McCandless’s website, informationisbeautiful.net. Find an infographic that you think is particularly interesting and effective. Share it with your class.

Lesson B TED TALKS

THE BEAUTY OF DATA VISUALIZATION



DAVID McCANDLESS Data journalist, TED speaker

David McCandless makes infographics—simple, elegant ways to understand complex information. Surprisingly, McCandless started designing without formal instruction in art or design.

McCandless started working as a computer programmer and then wrote articles for magazines and websites for about 20 years. However, when he started designing infographics, he realized that he already knew a lot about design. As he explains, “I was sensitive to the ideas of grids and space and alignment and typography.” Years of looking at media, like magazine articles and websites, gave McCandless a basic understanding of good design. His book *Information Is Beautiful* includes infographics about dozens of topics, from politics to pop music.

grids: *n.* patterns of straight lines that cross each other to make squares

alignment: *n.* arrangement in a straight line, or in appropriate positions

typography: *n.* the style and appearance of printed letters and words

In this lesson, you are going to watch segments from McCandless's TED Talk. Use the information above about McCandless to answer these questions.

1. Given McCandless's present job, what is surprising about his education and work experience?

He does not have formal training in design or in computer graphics.

2. How do you think the jobs of a journalist and an infographic designer are similar?

Both a journalist and an infographic designer have to research and gather information, and then convey the information to others in a meaningful way

3. What lessons did McCandless learn about designing infographics from his previous jobs?

He had looked at many visuals during his research as a journalist, which gave him a basic understanding of good design.

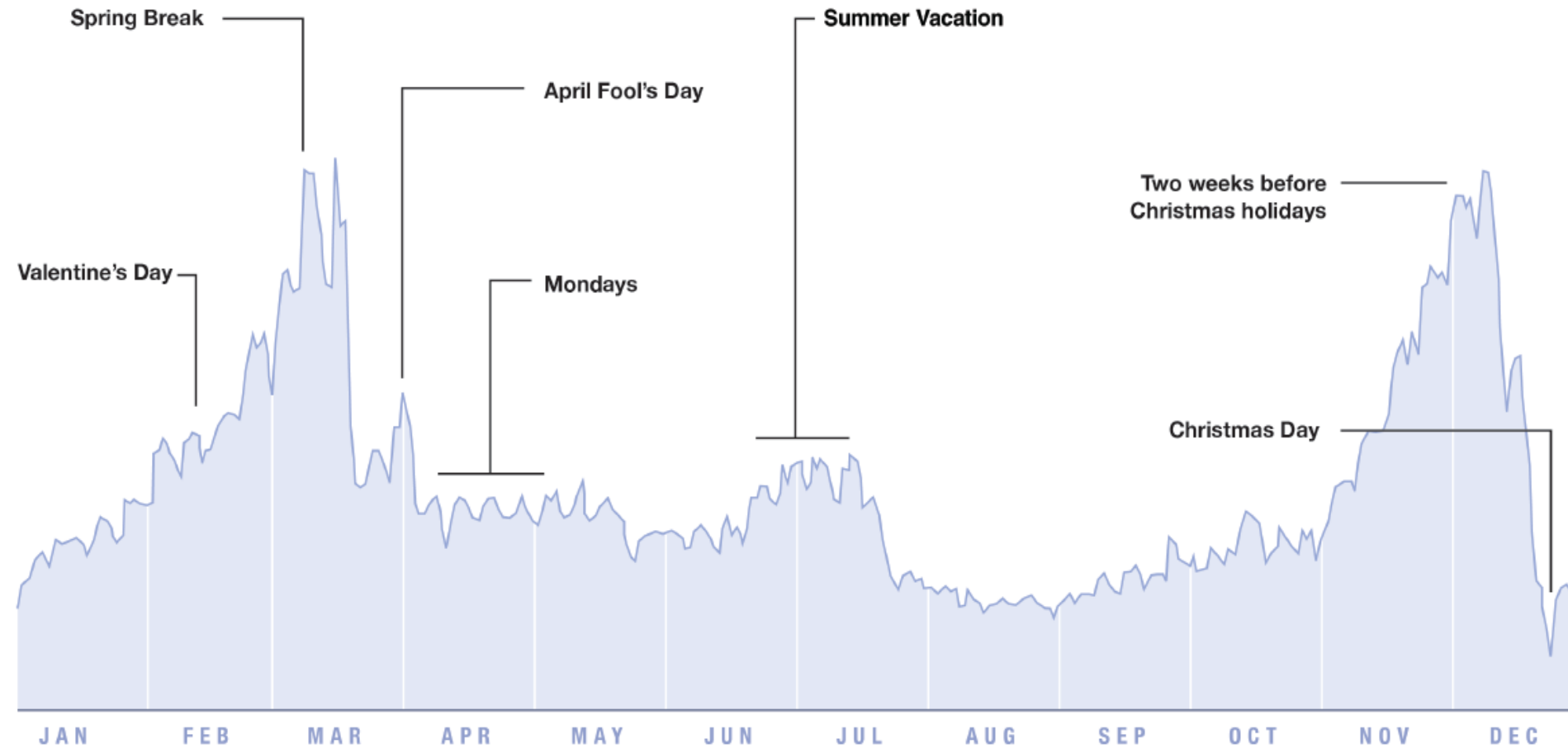
TED TALKS

McCandless's **idea worth spreading** is that visualizing data into beautiful designs can help us understand complex information.



Peak Break-Up Times

According to Facebook status updates



PART 1

MINING INFORMATION

PREVIEWING

- A.** Look at the infographic above. What does break-up mean? Why do you think the number of posts about break-ups is higher or lower at certain times of the year? Discuss with a partner.

A “break-up” refers to the end of a romantic relationship. There are likely easier times to break up with someone, such as before Spring Break, than others, like Christmas Day.

- B. Read this excerpt from the talk. What words do you think are missing? Fill in each blank and share your ideas with a partner.**

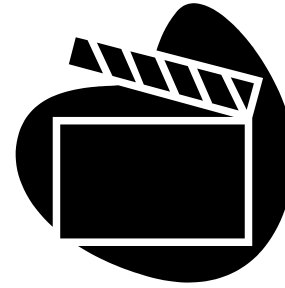
Guesses will vary. Actual answers are:

“[Data] is a really fertile medium, and it feels like visualizations, infographics, data visualizations, they feel like flowers blooming from this medium. But if you look at it directly, it’s just a lot of numbers and disconnected facts. But if you start working with it and playing with it in a certain way, interesting things can appear and different patterns can be revealed.”

fertile: *adj.* full of possibilities; when used to describe land: able to produce plants

medium: *n.* a way of communicating with people

blooming: *v.* used to describe flowers: opening



GETTING THE MAIN IDEA

Watch (▶) the first segment of the talk and check your answers to the Previewing exercises. Then answer the questions below.

1. How did McCandless and Lee Byron collect the data for the infographic on page 144?

They took data from 10,000 Facebook status updates about break-ups.

2. What does McCandless do to find interesting things from this “titanic amount of data?”

He says they ask the right kinds of questions and work the data in the right kind of way. He means that he asks specific questions to find connections and patterns that focus his data search.

CRITICAL THINKING

1. **Reflecting.** Do you think the infographic tells the whole story about when people might break up with each other? Why or why not? Write your ideas and then discuss with a partner.

Answers will vary.

2. **Analyzing.** People say “data is the new oil,” but McCandless changes it to “data is the new soil.” How is data like oil? How is it like soil? Discuss with a partner.

Data is like oil because it is something humans have come to depend on and to mine. Data is like soil because it provides a platform from which to cultivate and grow interesting ideas. We are sowing this “soil” by finding the connections, patterns, and stories in it that McCandless talks about.

PART 2

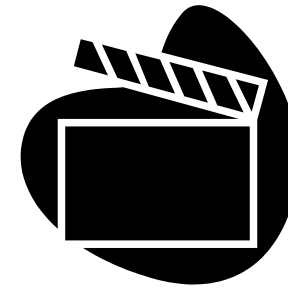
INTERACTING WITH DATA

UNDERSTANDING MAIN IDEAS

Read the quote below and choose the statement that best describes what McCandless means.

“ . . . visualizing information like this is a form of knowledge compression. It’s a way of squeezing an enormous amount of information and understanding into a small space.”

- a. It’s difficult to see the connections between different types of data.
- ☒ b. Infographics let us represent a lot of data in pictures.
- c. When you create an infographic, you have to do a lot of research.



UNDERSTANDING KEY DETAILS

A. The statements below are about the “Snake Oil” infographic in McCandless’s talk. Watch () the next segment of McCandless’s talk and write *T* for True and *F* for False.

- F* 1. There is less evidence that the supplements at the top are effective.
- F* 2. McCandless found all of the information for this infographic from one study.
- T* 3. It takes a long time to update the infographic with new information.
- T* 4. With the interactive version, you can choose to see only certain supplements.

B. Read this excerpt below from McCandless's talk. Match each bolded word with the correct definition.

“[I]nformation design is about solving information problems. It feels like we have a lot of information problems in our society at the moment, from the overload and the saturation to the **breakdown** of trust and **reliability** and runaway **skepticism** and lack of **transparency**, or even just interestingness. I mean, I find information just too interesting. It has a magnetic quality that draws me in.”

 d 1. breakdown

 b 2. reliability

 a 3. skepticism

 c 4. transparency

a. doubt that something is true

b. ability to be trusted

c. openness

d. a situation in which something is failing

CRITICAL THINKING

1. **Interpreting.** McCandless says: “Let the dataset change your mindset.” Explain this idea in your own words. How does the “Snake Oil” infographic illustrate this idea?

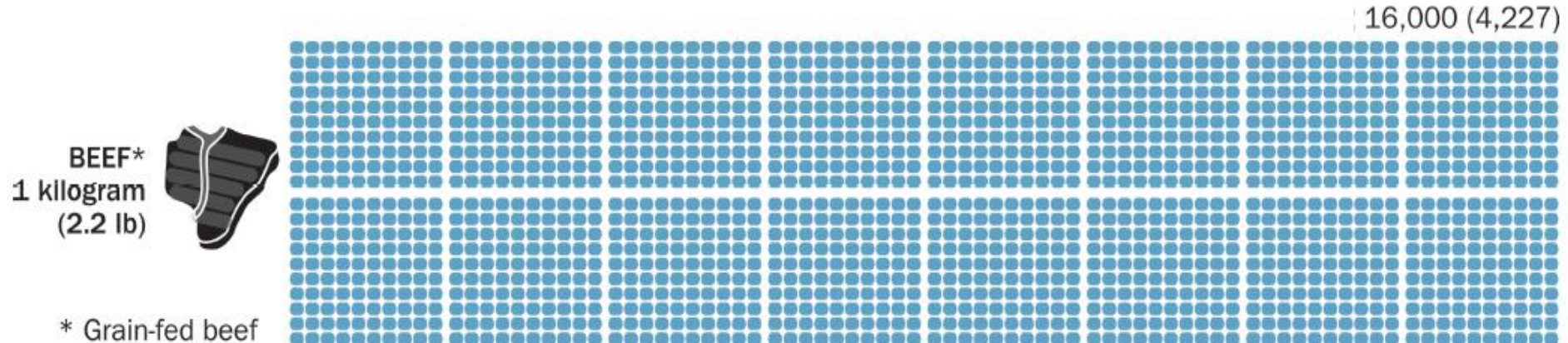
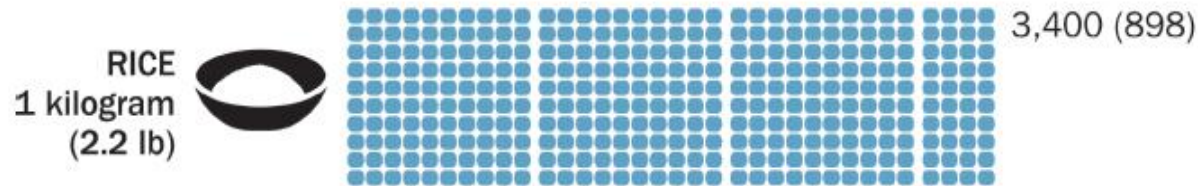
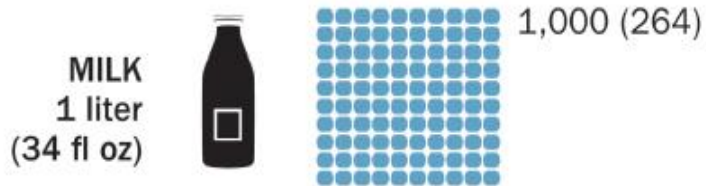
Answers will vary. He means that the information can change our thinking because it can show us a more well-rounded and representative picture by helping us see connections and patterns that we might not otherwise have noticed.

2. **Reflecting.** In your opinion, which infographic from McCandless’s TED Talk is the most effective? Why? *Answers will vary.*

EXPLORE MORE

Watch McCandless’s full TED Talk at TED.com. What did you learn from the other infographics he shows? Share your information with the class.

Project



Thirsty World

The chart shows the number of liters of water required for production of everyday food items.

● = 10 liters (2.6 gallons)

Project

- A. Work with a partner. You are going to research and present an infographic. Follow the steps below:**
1. Find a topic to investigate. You can use one of the topics from this book.
 2. Research your topic to find relevant and interesting data.
 3. Plan the most important pieces of data you want to show visually. For example, if you have data representing a number of categories, you could present the information as a chart like the one above.
- B. Use the data you collected above to create an infographic. Use the information you learned in this unit about well-designed infographics.**
- C. Use your information to create a short presentation. Use your infographic to explain your information.**

Project

EXPLORE MORE

Watch Aaron Koblin's TED Talk "Visualizing ourselves . . . with crowd-sourced data." Choose one of the infographics Koblin created and explain it to your class.

Part 1

. . . So, I've been working as a data journalist for about a year, and I keep hearing a phrase all the time, which is this: "Data is the new oil." Data is the kind of ubiquitous resource that we can shape to provide new innovations and new insights, and it's all around us, and it can be mined very easily. It's not a particularly great metaphor in these times, especially if you live around the Gulf of Mexico, but I would, perhaps, adapt this metaphor slightly, and I would say that data is the new soil. Because for me, it feels like a fertile, creative medium. Over the years, online, we've laid down a huge amount of information and data, and we irrigate it with networks and connectivity, and it's been worked and tilled by unpaid workers and governments. And, all right, I'm kind of milking the metaphor a little bit. But it's a really fertile medium, and it feels like visualizations, infographics, data visualizations, they feel like flowers blooming from this medium. But if you look at it directly, it's just a lot of numbers and disconnected facts. But if you start working with it and playing with it in a certain way, interesting things can appear and different patterns can be revealed.

Let me show you this. Can you guess what this data set is? What rises twice a year, once in Easter and then two weeks before Christmas, has a mini peak every Monday, and then flattens out over the summer? I'll take answers.

[Audience: Chocolate.] David McCandless: Chocolate. You might want to get some chocolate in. Any other guesses? [Audience: Shopping.] DM: Shopping. Yeah, retail therapy might help. [Audience: Sick leave.] DM: Sick leave. Yeah, you'll definitely want to take some time off. Shall we see?

So, the information guru Lee Byron and myself, we scraped 10,000 status Facebook updates for the phrase "break-up" and "broken-up" and this is the pattern we found —people clearing out for Spring Break, [Laughter] coming out of very bad weekends on a Monday, being single over the summer, and then the lowest day of the year, of course: Christmas Day. Who would do that? So there's a titanic amount of data out there now, unprecedented. But if you ask the right kind of question, or you work it in the right kind of way, interesting things can emerge. . . .

Part 2

We need relative figures that are connected to other data so that we can see a fuller picture, and then that can lead to us changing our perspective. As Hans Rosling, the master, my master, said, “Let the dataset change your mindset.” And if it can do that, maybe it can also change your behavior.

Take a look at this one. I’m a bit of a health nut. I love taking supplements and being fit, but I can never understand what’s going on in terms of evidence. There’s always conflicting evidence. Should I take vitamin C? Should I be taking wheatgrass? This is a visualization of all the evidence for nutritional supplements. This kind of diagram is called a balloon race. So the higher up the image, the more evidence there is for each supplement. And the bubbles correspond to popularity as regards to Google hits. So you can immediately apprehend the relationship between efficacy and popularity, but you can also, if you grade the evidence, do a “worth it” line. So supplements above this line are worth investigating, but only for the conditions listed below, and then the supplements below the line are perhaps not worth investigating.

Now this image constitutes a huge amount of work. We scraped like 1,000 studies from PubMed, the biomedical database, and we compiled them and graded them all. And it was incredibly frustrating for me because I had a book of 250 visualizations to do for my book, and I spent a month doing this, and I only filled two pages. But what it points to is that visualizing information like this is a form of knowledge compression. It's a way of squeezing an enormous amount of information and understanding into a small space. And once you've curated that data, and once you've cleaned that data, and once it's there, you can do cool stuff like this.

So I converted this into an interactive app, so I can now generate this application online—this is the visualization online—and I can say, “Yeah, brilliant.” So it spawns itself. And then I can say, “Well, just show me the stuff that affects heart health.” So let’s filter that out. So heart is filtered out, so I can see if I’m curious about that. I think, “No, no. I don’t want to take any synthetics, I just want to see plants and—just show me herbs and plants. I’ve got all the natural ingredients.” Now this app is spawning itself from the data. The data is all stored in a Google Doc, and it’s literally generating itself from that data. So the data is now alive; this is a living image, and I can update it in a second. New evidence comes out. I just change a row on a spreadsheet. Doosh! Again, the image recreates itself. So it’s cool. It’s kind of living. . . .

So, just to wrap up, I wanted to say that it feels to me that design is about solving problems and providing elegant solutions, and information design is about solving information problems. It feels like we have a lot of information problems in our society at the moment, from the overload and the saturation to the breakdown of trust and reliability and runaway skepticism and lack of transparency, or even just interestingness. I mean, I find information just too interesting. It has a magnetic quality that draws me in. . . .